

1. Write a shell script to show various system configuration like currently logged user and his logname, your current shell, home directory, operating system type, current path setting, current working directory, show currently logged number of users, show memory information, Hard disk information like size of hard-disk, cache memory, model etc, and file system mounted.

```
#!/bin/bash
echo "currently logged in user :$(whoami)"
echo "Logname : $(logname 2>/dev/null || echo $USER)"
echo "Current shell :$SHELL"
echo "home directory :$HOME"
echo "Operating system type :$(uname -o 2>/dev/null || uname-s)"
echo "Current path setting :$PATH"
echo "Current working directory :$(pwd)"
echo "Logged in user count :$(who | wc -l)"
echo "Memory info"
free -h
echo "CPU info"
lscpu
echo "Disk info"
lsblk
echo "disk usage"
df -h
echo "Mounted file system"
mount
```

2. write a shell script to find the factorial of given integer

```
#!/bin/bash
echo -n "Enter a num"
read num
fact=1
for ((i=1;i<=num;i++))
do
fact=$((fact*i))
done
echo "$fact"
```

1. Write a shell script to add user and password on Linux system.

```
#!/bin/bash
echo -n "enter username: "
read username
if id "$username" &>/dev/null;
then
echo "user already exists"
exit 1
fi
sudo useradd -m "$username"
echo "set password"
sudo passwd "$username"
echo "user $username created"
```

2. Write a shell script to print last login details.

```
#!/bin/bash
echo "last login details of current user"
last $USER
echo "last login details of all user"
last
echo "current logged in user name:"
who
```

3. Write a shell script to upgrade and cleans the system automatically instead of doing it manually.

```
#!/bin/bash
echo "Updating system"
sudo apt update
echo "Upgrading packages"
sudo apt upgrade -y
echo "Removing unnecessary files"
sudo apt autoremove -y
echo "Cleaning cache"
sudo apt autoclean
```

Write script to perform following String Operation

- Find Length of String
- Find and Replace String
- To Concatenate String
- Reversing the string

```
#!/bin/bash
echo "Enter a string:"
read str
echo "Length of string: ${#str}"
echo "Enter word to find:"
read find
echo "Enter word to replace:"
read replace
echo "After replacement: ${str/$find/$replace}"
echo "Enter another string to concatenate:"
read str2
echo "Concatenated string: $str $str2"
echo "Reversed string: $(echo "$str" | rev)"
```

- a. Write a shell script to delete all log files present inside your var/log directory
- b. Write script to check if user is root user
- c. Write script to print number of files and folders
- d. Write Script to Create Directories

```
#!/bin/bash
echo "Deleting all .log files in /var/log..."
sudo rm -f /var/log/*.log
```

```
echo "check if user root or not"
if [ "$EUID" -eq 0 ]
then
echo "You are root user"
else
echo "You are NOT root user"
fi
```

```
echo "number of folders and files in directory"
echo "Enter directory path:"
read dir
files=$(find "$dir" -type f | wc -l)
folders=$(find "$dir" -type d | wc -l)
echo "Number of files: $files"
echo "Number of folders: $folders"
```

```
echo "create directories"
echo "Enter directory name:"
read directory
mkdir "$directory"
echo "Directory created"
```

Write Shell Script to find maximum of three numbers, read number from user

```
#!/bin/bash
echo "Enter first number:"
read a
echo "Enter second number:"
read b
echo "Enter third number:"
read c
if [ $a -ge $b ] && [ $a -ge $c ]
then
max=$a
elif [ $b -ge $a ] && [ $b -ge $c ]
then
max=$b
else
max=$c
fi
echo "Maximum number is: $max"
```

Write a script that accepts the hostname and IP address as command-line arguments and adds them to the /etc/hosts file.

Write a shell script that delete all lines containing a specified word

Write Bash script to display name and a greeting message with some information

```
#!/bin/bash
ip=$1
host=$2
echo "Adding to /etc/hosts"
echo "$ip $host" | sudo tee -a /etc/hosts
echo "Entry added: $ip $host"
```

how to run - ./script.sh 192.168.1.10 myserver

```
#!/bin/bash
echo "Enter filename:"
read file
echo "Enter word to delete:"
read word
sed -i "/$word/d" "$file"
echo "Lines containing '$word' deleted"
```

how to run – normal later in filename write - jash.txt

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Hello, $name!"
echo "Welcome to the system"
echo "Today is: $(date)"
echo "Your username is: $USER"
echo "Your home directory is: $HOME"
```

Write Menu Drive program using Shell script to Perform Following operation like Addition, Subtraction , multiplication, Division, remainder

```
#!/bin/bash
echo "Choose operation:"
echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"
echo "5. Remainder"
read choice
echo "Enter first number:"
read a
echo "Enter second number:"
read b
case $choice in
1) echo "Result:  $$(a + b)$ " ;;
2) echo "Result:  $$(a - b)$ " ;;
3) echo "Result:  $$(a * b)$ " ;;
4) echo "Result:  $$(a / b)$ " ;;
5) echo "Result:  $$(a \% b)$ " ;;
*) echo "Invalid choice" ;;
esac
```

Write shell Script to display Following information using environment variables like

SHELL,USER,PWD,PATH,LANG,HOME,BASH_VERSION,BASH_VERSI

NFO
HOSTNAME,UID etc

```
#!/bin/bash
echo "User          : $USER"
echo "Shell          : $SHELL"
echo "Current Dir     : $PWD"
echo "Path            : $PATH"
echo "Language         : $LANG"
echo "Home Directory   : $HOME"
echo "Bash Version     : $BASH_VERSION"
echo "Bash Info        : ${BASH_VERSINFO[@]}"
echo "Hostname         : $HOSTNAME"
echo "User ID          : $UID"
```

Write Shell script using for loop to generate fibonacci Series till the limit specified

by user

```
#!/bin/bash
echo -n "Enter limit:"
read limit
a=0
b=1
echo "Fibonacci Series:"
for (( i=1; i<=limit; i++ ))
do
echo "$a "
c=$((a + b))
a=$b
b=$c
done
echo
```


Write shell script to perform Following String Operations

1. Checks if the given string operand size is non-zero
2. Checks if the given string operand size is zero
3. Checks if the value of two operands are equal
4. Checks if the value of two operands are not equal
5. Checks if str is not the empty string; if it is empty (Same as 2)

```
#!/bin/bash
echo "Enter first string:"
read str1
echo "Enter second string:"
read str2

if [ -n "$str1" ]
then
echo "String1 is NOT empty"
else
echo "String1 is empty"
fi

if [ -z "$str1" ]
then
echo "String1 is empty"
else
echo "String1 is NOT empty"
fi

if [ "$str1" = "$str2" ]
then
echo "Both strings are equal"
else
echo "Strings are not equal"
fi

if [ "$str1" != "$str2" ]
then
echo "Strings are not equal"
else
echo "Strings are equal"
fi
```

Write Shell Script to perform Following Operations

1. Checks if file is a block special file
2. Checks if file is a character special file
3. Checks if file is a directory
4. Checks if file is an ordinary file as opposed to a directory or specialfile
5. Checks if file has its set group ID (SGID) bit set
6. Checks if file has its sticky bit set
7. Checks if file is a named pipe
8. Checks if file descriptor is open and associated with a terminal

```
#!/bin/bash
```

```
echo "Enter file name or path:"
```

```
read file
```

```
# 1. Block special file
```

```
if [ -b "$file" ]
```

```
then
```

```
echo "It is a block special file"
```

```
fi
```

```
# 2. Character special file
```

```
if [ -c "$file" ]
```

```
then
```

```
echo "It is a character special file"
```

```
fi
```

```
# 3. Directory
```

```
if [ -d "$file" ]
```

```
then
```

```
echo "It is a directory"
```

```
fi
```

```
# 4. Ordinary file (regular file)
```

```
if [ -f "$file" ]
```

```
then
```

```
echo "It is a regular file"
```

```
fi
```

```
# 5. SGID bit set
if [ -g "$file" ]
then
echo "SGID bit is set"
fi
```

```
# 6. Sticky bit set
if [ -k "$file" ]
then
echo "Sticky bit is set"
fi
```

```
# 7. Named pipe
if [ -p "$file" ]
echo "It is a named pipe"
fi
```

```
# 8. File descriptor is terminal
if [ -t 1 ]
then
echo "Output is a terminal"
else
echo "Output is not a terminal"
fi
```

Write bash script to find sum of square of n numbers, read n from user

```
#!/bin/bash
echo "Enter how many numbers (n):"
read n
sum=0
for (( i=1; i<=n; i++ ))
do
echo "Enter number $i:"
read num
sum=$((sum + num*num))
done
echo "Sum of squares = $sum"
```

Write shell script to demonstrate Command Line Argument

i.Positional Parameters

ii.Using Flag

iii.Using Loops with \$@ – Loop Constructs

iv. Use of \$#,\$*,\$\$,\$? And S!

```
#!/bin/bash
```

```
echo "==== Positional Parameters ====="
```

```
echo "Script name: $0"
```

```
echo "1st argument: $1"
```

```
echo "2nd argument: $2"
```

```
echo ""
```

```
echo "==== Special Variables ====="
```

```
echo "Number of arguments (#): $#"
```

```
echo "All arguments (*): $*"
```

```
echo "Process ID ($$): $$"
```

```
echo "Last command status ($?): $?"
```

```
echo "Last background process ($!): $!"
```

```
echo ""
```

```
echo "==== Using Flag ====="
```

```
if [ "$1" = "-a" ]
```

```
then
```

```
    echo "Option A selected"
```

```
elif [ "$1" = "-b" ]
```

```
then
```

```
    echo "Option B selected"
```

```
else
```

```
    echo "No valid flag"
```

```
fi
```

```
echo ""
```

```
echo "==== Loop using \@ ====="
```

```
for arg in "$@"
```

```
do
```

```
    echo "Argument: $arg"
```

```
done
```

Write Bash script to find whether character is vowel , consonant, Special Character or Digit use switch case

```
#!/bin/bash
echo "Enter a character:"
read ch
case $ch in
[aeiouAEIOU]) echo "Vowel" ;;
[a-zA-Z]) echo "Consonant" ;;
[0-9]) echo "Digit" ;;
*) echo "Special Character" ;;
esac
```

Write shell script to demonstrate Sort command with different sort command
Option sort -b,sort -r,sort -o,sort -n,sort -M,sort -u,sort -k,sort -t SEP

```
#!/bin/bash
```

```
echo "Creating sample file..."
```

```
cat > data.txt <<EOF
```

```
    apple
```

```
banana
```

```
Orange
```

```
apple
```

```
10
```

```
2
```

```
Jan
```

```
Feb
```

```
Mar
```

```
EOF
```

```
echo ""
```

```
echo "Original File:"
```

```
cat data.txt
```

```
echo ""
```

```
echo "1. sort -b (ignore leading spaces):"
```

```
sort -b data.txt
```

```
echo ""
```

```
echo "2. sort -r (reverse order):"
```

```
sort -r data.txt
```

```
echo ""
```

```
echo "3. sort -n (numeric sort):"
```

```
sort -n data.txt
```

```
echo ""
```

```
echo "4. sort -M (month sort):"
```

```
sort -M data.txt
```

```
echo ""
```

```
echo "5. sort -u (unique values):"  
sort -u data.txt
```

```
echo ""  
echo "6. sort -o (output to file):"  
sort data.txt -o sorted.txt  
echo "Sorted data saved in sorted.txt"  
cat sorted.txt
```

```
echo ""  
echo "7. sort -k (sort by column):"  
echo -e "3 Ram\n1 Shyam\n2 Amit" > col.txt  
sort -k1 col.txt
```

```
echo ""  
echo "8. sort -t (delimiter):"  
echo -e "1:Ram\n3:Amit\n2:Shyam" > sep.txt  
sort -t ":" -k1 sep.txt
```

Using WC Command Perform the following

- A. Display Line word and Character count in a file
- B. Count Number of Lines in a File
- C. Count Number of Words in a File
- D. Count Number of Characters in a File
- E. Count Number of Bytes in a File
- F. Display Length of Longest Line in File

```
#!/bin/bash
echo "Enter filename:"
read file
echo "A. Lines, Words, Characters:"
wc "$file"
echo "B. Number of Lines:"
wc -l "$file"
echo "C. Number of Words:"
wc -w "$file"
echo "D. Number of Characters:"
wc -m "$file"
echo "E. Number of Bytes:"
wc -c "$file"
echo "F. Length of Longest Line:"
wc -L "$file"
```

Write a shell program to check if a given string is a palindrome or not

```
#!/bin/bash
echo "Enter a string:"
read str
revstr=$(echo "$str" | rev)
if [ "$str" = "$revstr" ]
then
echo "It is a palindrome"
else
echo "It is not a palindrome"
fi
```


Write script to Perform Following Operation

1. Print ipaddress, Subnetmask, gateway and DNS
2. Check the Number of Logged-in User
3. Check Operating System Informaton
4. Check System Memory usage
5. Check System's Disk Usage
6. Check system Network Information
7. Check system Architecture

```
#!/bin/bash
# 1. IP Address, Gateway, DNS
echo ""
echo "1. Network Details:"
echo "IP Address:"
hostname -I
echo "Gateway:"
ip route | grep default
echo "DNS:"
cat /etc/resolv.conf | grep nameserver
# 2. Logged-in users
echo ""
echo "2. Logged-in Users:"
who | wc -l
# 3. Operating System Info
echo ""
echo "3. Operating System:"
uname -a
# 4. Memory Usage
echo ""
echo "4. Memory Usage:"
free -h
# 5. Disk Usage
echo ""
echo "5. Disk Usage:"
df -h
# 6. Network Information
echo ""
echo "6. Network Info:"
ip a
# 7. System Architecture
```

```
echo ""  
echo "7. System Architecture:"  
uname -m
```

Write shell script to find reverse of 5 digit number using while loop

```
#!/bin/bash
echo "Enter a 5-digit number:"
read num
rev=0
while [ $num -gt 0 ]
do
rem=$((num % 10))
rev=$((rev * 10 + rem))
num=$((num / 10))
done
echo "Reversed number: $rev"
```

Write shell script to generate prime number between limit specified by user
using
for loop

```
#!/bin/bash
echo "Enter limit:"
read n
echo "Prime numbers up to $n are:"
for (( i=2; i<=n; i++ ))
do
    isPrime=1

    for (( j=2; j<i; j++ ))
    do
        if [ $((i % j)) -eq 0 ]
        then
            isPrime=0
            break
        fi
    done
    if [ $isPrime -eq 1 ]
    then
        echo -n "$i "
    fi
done
echo
```

Write shell script to Generate sum of N numbers. Read N from user and until loop

```
#!/bin/bash
echo "Enter value of N:"
read n
i=1
sum=0
until [ $i -gt $n ]
do
sum=$((sum + i))
i=$((i + 1))
done
echo "Sum of first $n numbers is: $sum"
```

Write shell script to display 4x4 matrix and read data from user from keyboard.
Use for Loop

```
#!/bin/bash
echo "Enter elements of 4x4 matrix:"
declare -a matrix
# Input
for (( i=0; i<4; i++ ))
do
    for (( j=0; j<4; j++ ))
    do
        echo -n "Enter element [i][j]: "
        read val
        index=$((i*4 + j))
        matrix[$index]=$val
    done
done

echo ""
echo "Matrix is:"

# Output
for (( i=0; i<4; i++ ))
do
    for (( j=0; j<4; j++ ))
    do
        index=$((i*4 + j))
        echo -n "${matrix[$index]} "
    done
    echo ""
done
```

Write script to find sum of cube of 5 digit number

```
#!/bin/bash
echo "Enter a 5-digit number:"
read num
sum=0
while [ $num -gt 0 ]
do
    rem=$((num % 10))      # get last digit
    cube=$((rem * rem * rem)) # cube of digit
    sum=$((sum + cube))    # add to sum
    num=$((num / 10))      # remove last digit
done
echo "Sum of cubes of digits: $sum"
```

Configuration of NFS File server and transfer files to a windows client.

1. `sudo apt install nfs-kernel-server`
2. `sudo systemctl status nfs-kernel-server`
3. `sudo mkdir /home/nfsshare1`
4. `sudo chown nobody:nogroup /home/nfsshare1`
5. `sudo vi /etc/exports`
(/home/nfsshare1 192.168.1.0(rw,sync,no_subtree_check))
6. `sudo systemctl restart nfs-kernel-server`
7. `sudo ufw status`
8. `sudo ufw allow from 192.168.1.0 to any port nfs`
9. `ufw reload`

Client side.

1. `sudo apt update`
2. `sudo apt install nfs-common`
3. `sudo mkdir /mnt/nfserver100`
4. `sudo mount 192.168.1.0:/home/nfsshare1 /mnt/nfserver100`

Client side

1. `df -h`

Configuration of Web Server.

Configuration of Samba Server

Configuration of FTP server and transfer files to demonstrate the working of the same.